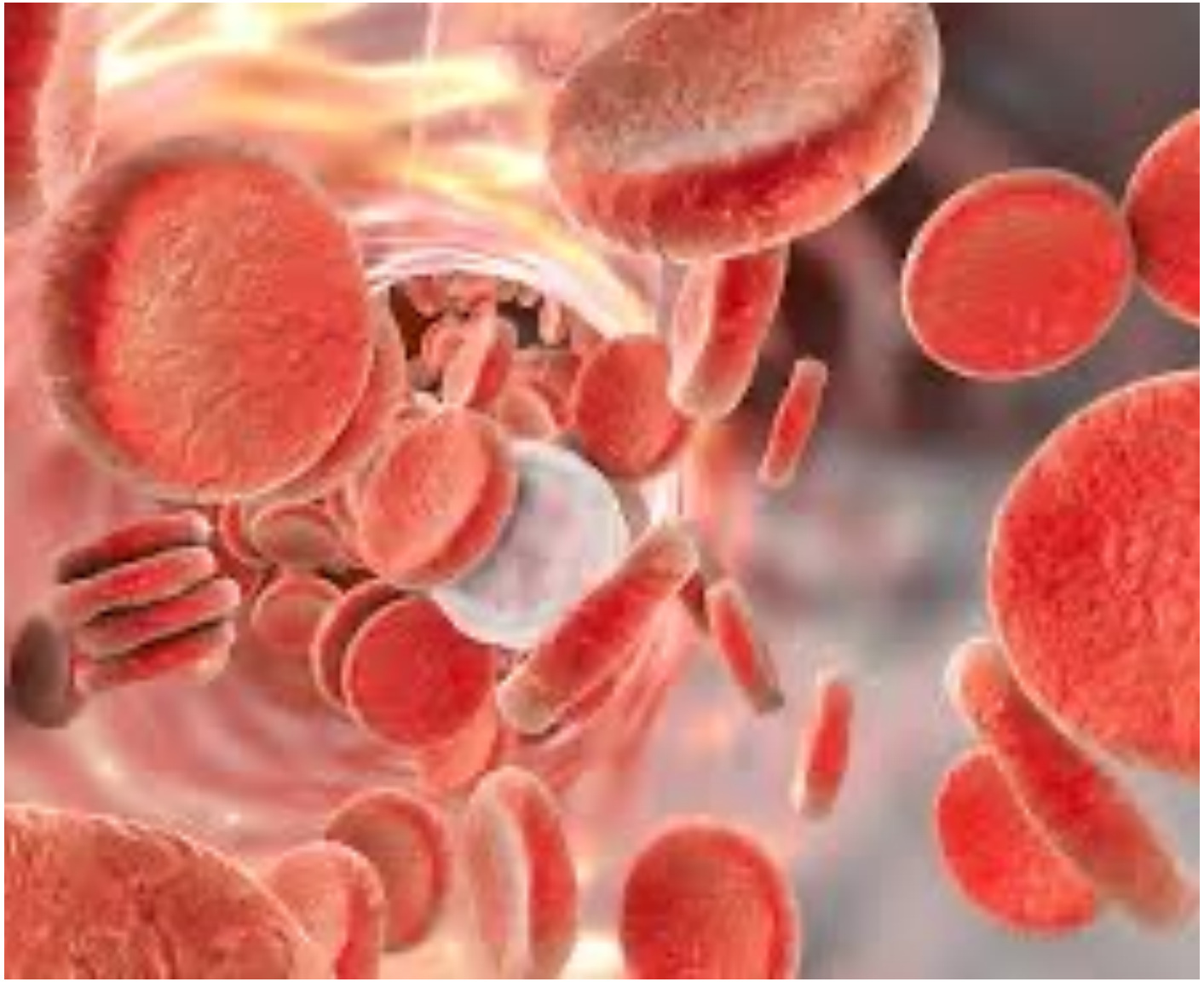
Guy's and St Thomas' NHS Foundation Trust

Royal Brompton & Harefield NHS Foundation Trust

Diffusion & Carriage of Blood Gases



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With thanks to Shauna Sheridan - Clinical Specialist Respiratory Physiotherapist

November 2024

LONDON METROPOLITAN UNIVERSITY

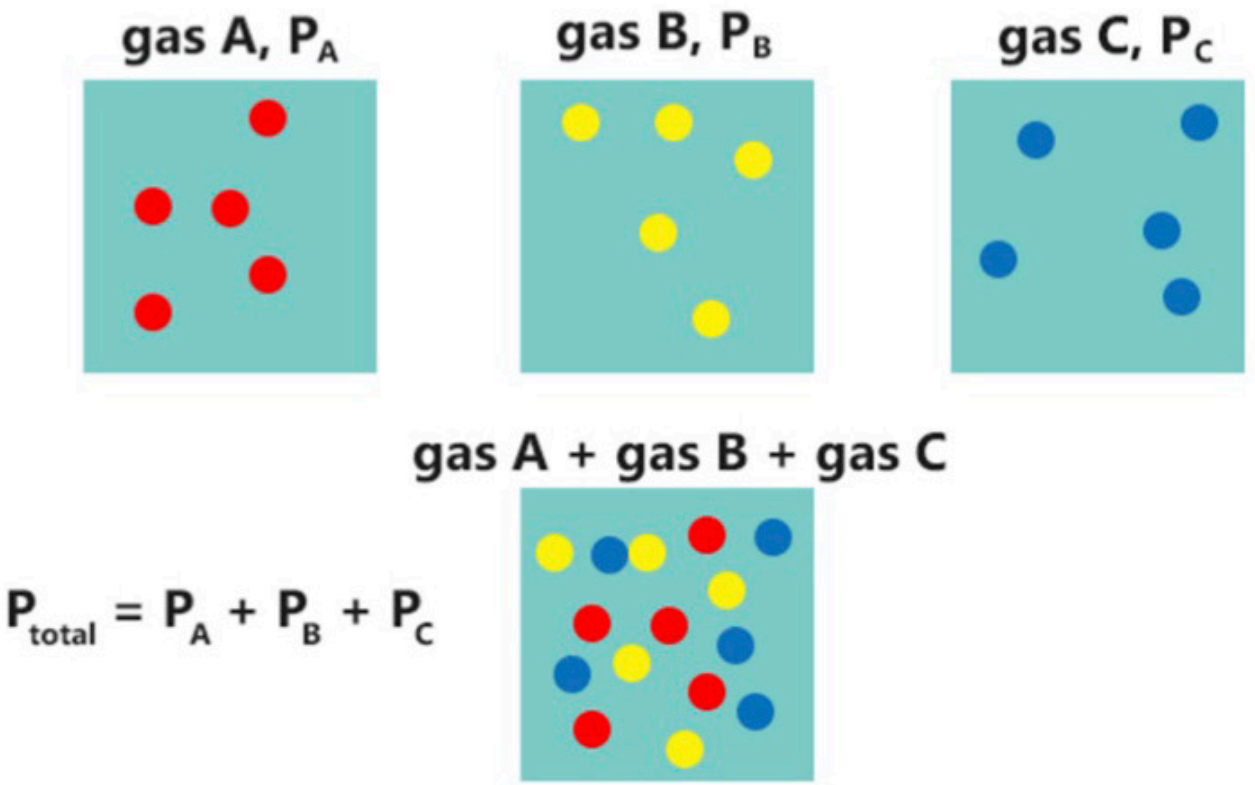
Learning Objectives

- Explain what is meant by partial pressure, fractional concentration, water vapour pressure and barometric pressure.
- Outline the composition of air and describe the effects of altering barometric pressure on partial pressures.
- Give the composition and partial pressures of gas in the airways and alveolar gas during breathing.
- Explain the factors affecting diffusion across the alveolar capillary membrane.
- Explain what is meant by oxygen partial pressure, oxygen capacity, content and saturation of blood and the factors that affect them.
- Describe the oxygen - haemoglobin dissociation curve.
- Explain how the affinity of the haemoglobin binding sites is affected by such things as temperature, pH and PCO_2 (Bohr shift)

Partial Pressure:

- Partial pressure : The pressure exerted by an individual gas in a mixture

- Daltons law expresses the fact that the total pressure of a mixture of ideal gases is equal to the sum of the partial pressures of the individual gases in a mixture



Fractional Concentration.

Water Vapour Pressure.

Barometric Pressure.

- Fractional concentration:

The composition of a single gas (gas A) , in the mixture (m) , can be represented in terms of its fractional concentration, the number of molecules of gas A divided by the number of all the molecules in the gas mixture

- Water vapour pressure:

The pressure exerted by molecules of water vapor in gaseous form (whether pure or mixed with other gases such as air)

- Barometric pressure:

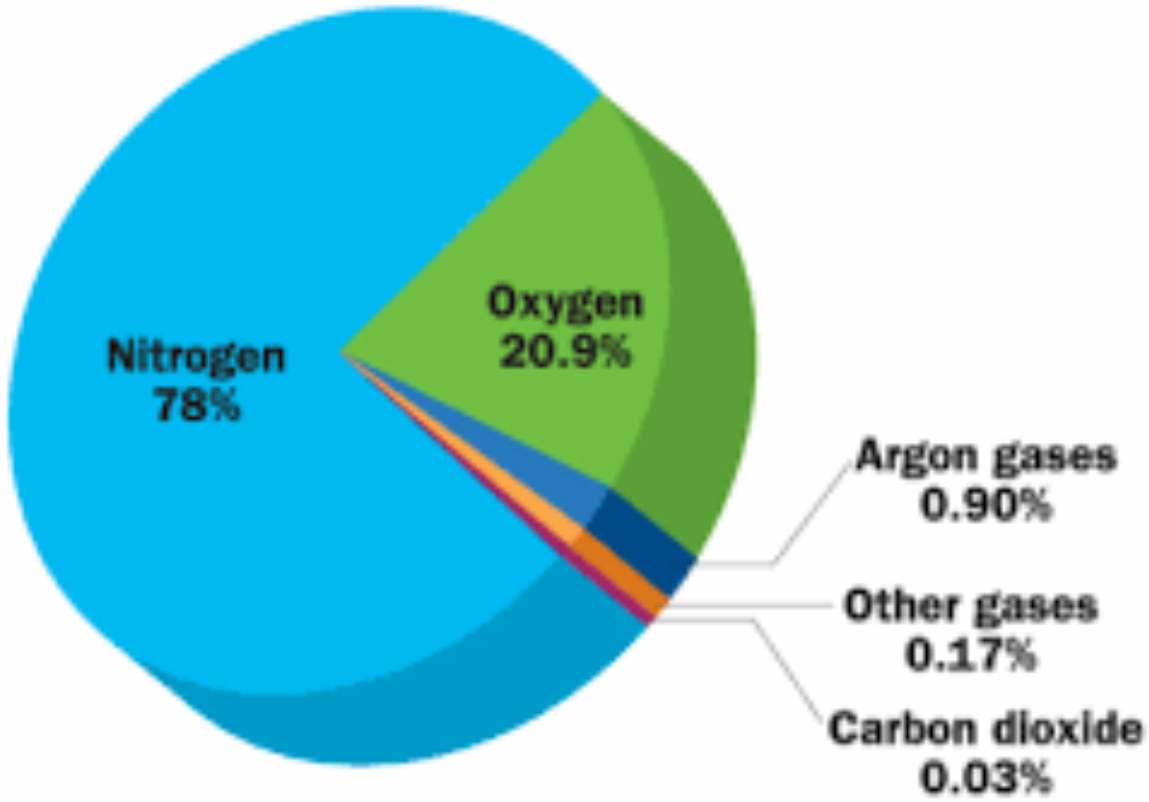
Can also be called atmospheric pressure/air pressure

760mmHg

Defined as the pressure within the atmosphere of the earth

Composition of air

- Nitrogen (N_2): 78%
  - N_2 keeps the alveoli open
  - It's inert and can't be absorbed.
- O_2 : 20.9%
- CO_2: 0.03%
- Other: 1.07%



Impact of atmospheric pressure change

At higher altitudes, atmospheric pr ↓ leading to a ↓ in the partial pressure of gases

The partial pressure of all gases will ↓ at higher altitudes because the overall pressure ↓

Pressure is essentially a measure of weight of ‘‘stuff’’ above you, in this case atmospheric gas

As you go higher, you are putting more of the atmosphere below you, leaving less of it above you

Therefore, the weight of gas pressing down on you decreases because there is less gas

Oxygen in the atmosphere remains constant (21%), therefore the only factor affecting partial pressure of oxygen is the atmospheric pressure change

Ficks Law:

- The rate of transfer of a gas through a sheet of tissue is proportional to:

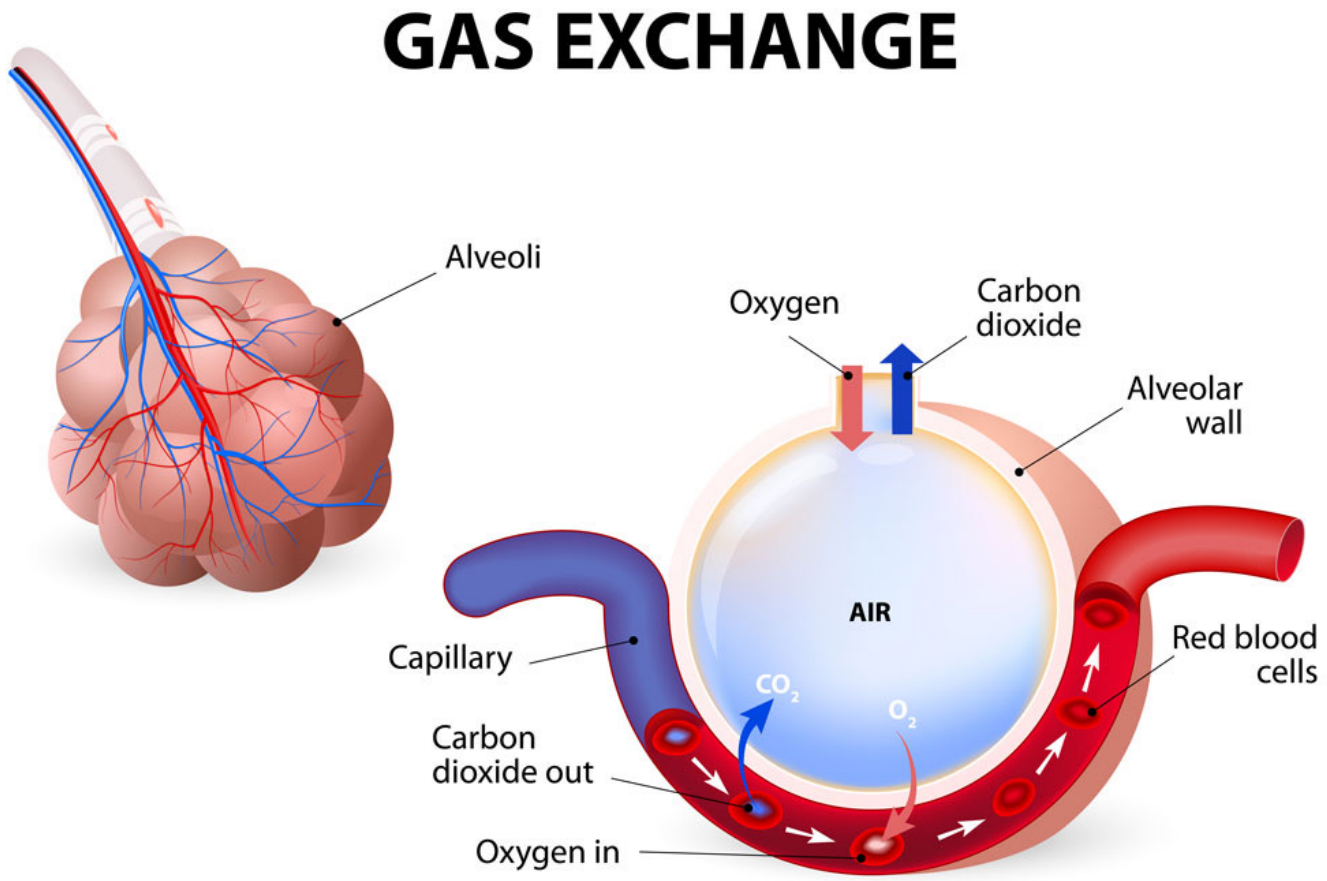
- Tissue area
- Difference in gas partial pressure between 2 sides
- Diffusion constant

*Inversely proportional to tissue thickness

Diffusion across alveolar membrane

Dependent on:

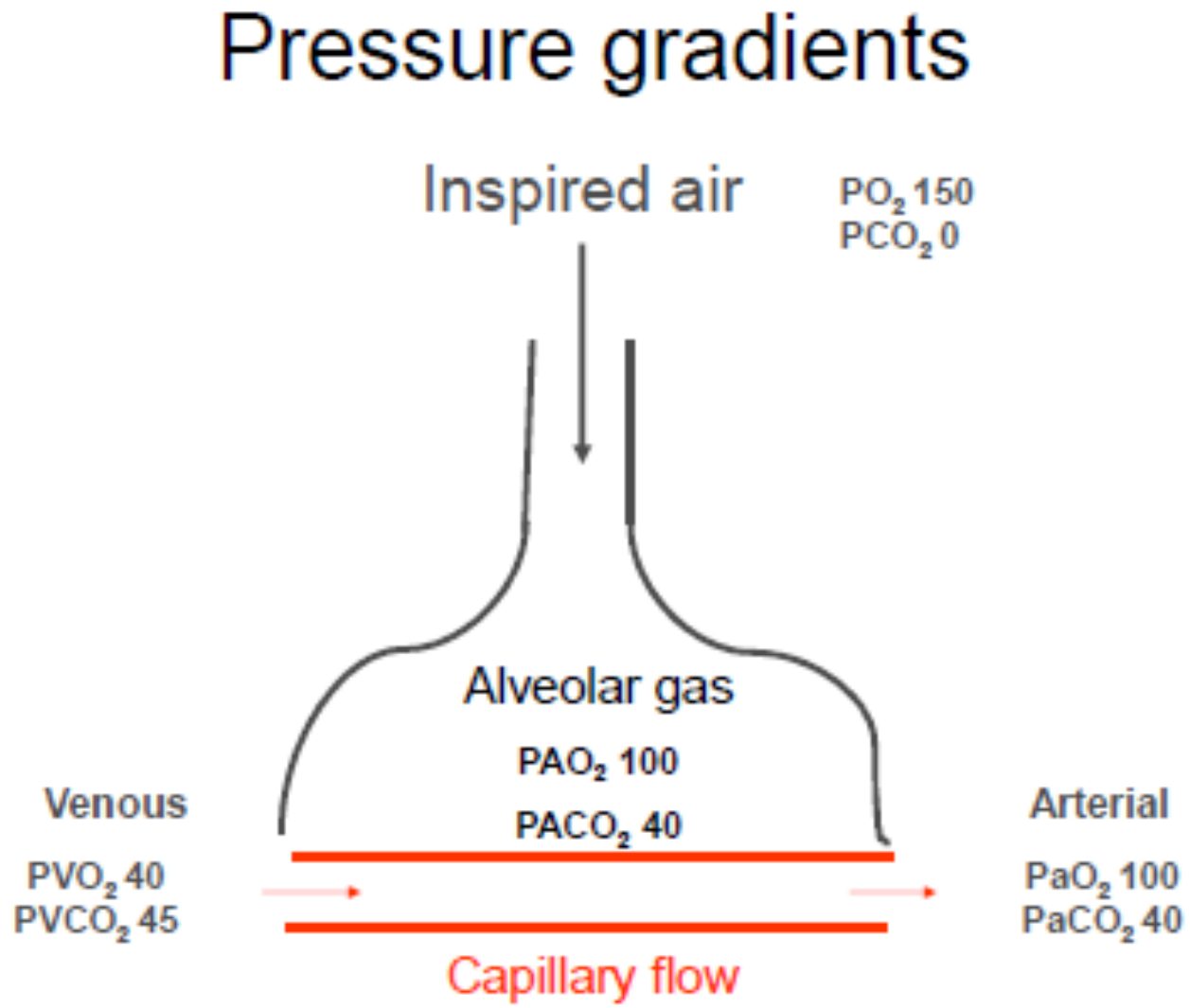
1. Concentration/ pressure gradient
2. Gas solubility
3. Thickness of alveolar membrane
4. Surface area of alveolar membrane
5. Ventilation/perfusion coupling



1. Concentration/pressure gradient

Partial pressure = amount of gas dissolved in the plasma

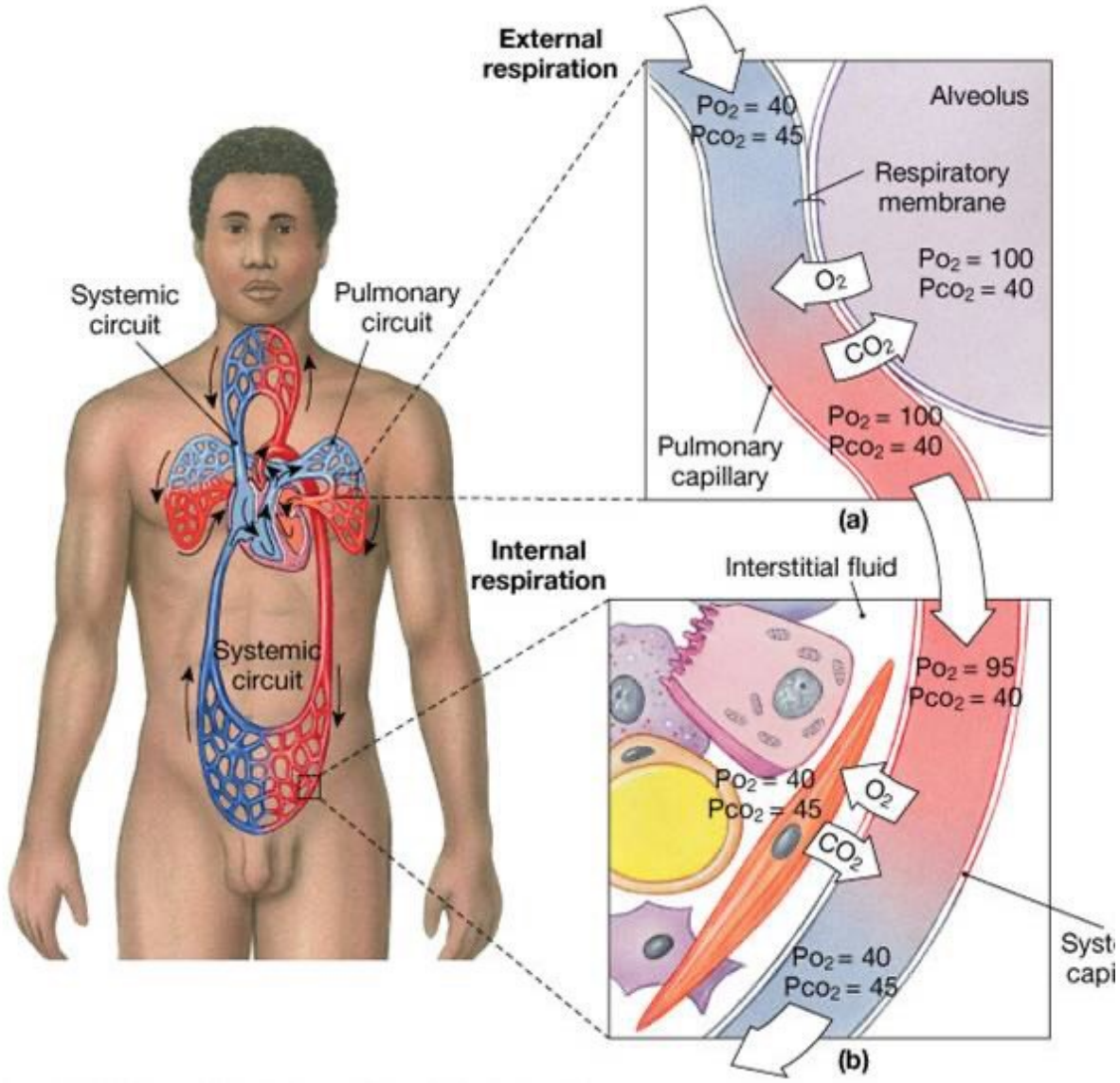
i.e: PaO_2 = amount of PO_2 dissolved in arterial blood



2. Gas solubility

- CO_2 is 20 times more soluble than O_2
- Equal amounts of CO_2 + O_2 diffuse across the respiratory membrane in the same time period

- Clinical relevance: In resp disease when diffusion is impaired, O_2 will be primarily affected as its less soluble

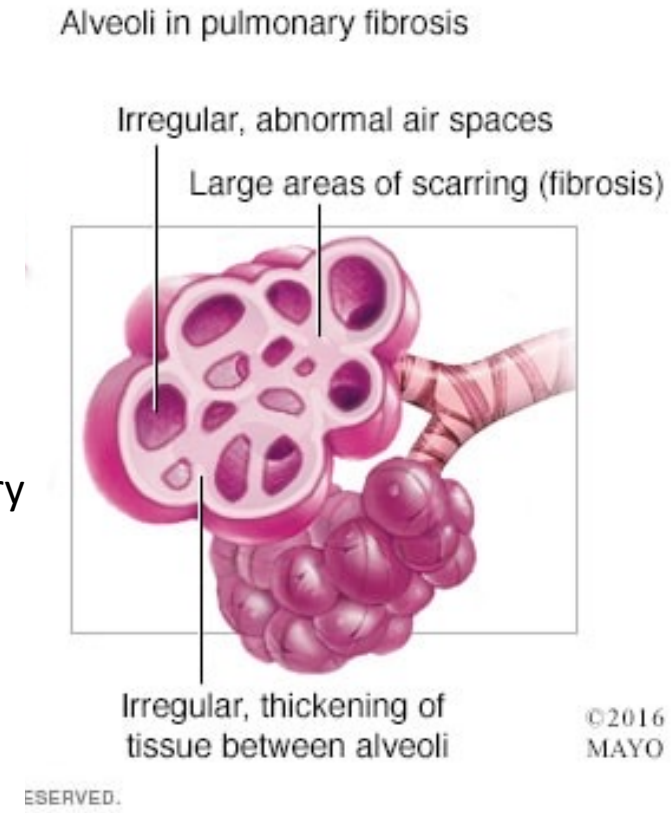


3. Thickness of membrane

- 0.5 - 1 thick
- Diffusion efficiency is highly dependent on distances involved
- Alveolar membrane is ultra thin, resulting in rapid and efficient gas exchange

Clinical significance:

- If membrane is thickened there is a greater distance between alveoli and capillary
- Diffusion slow and/or impaired
- Leading to hypoxia
- Thickening may occur due to inflammation, infection or fibrosis

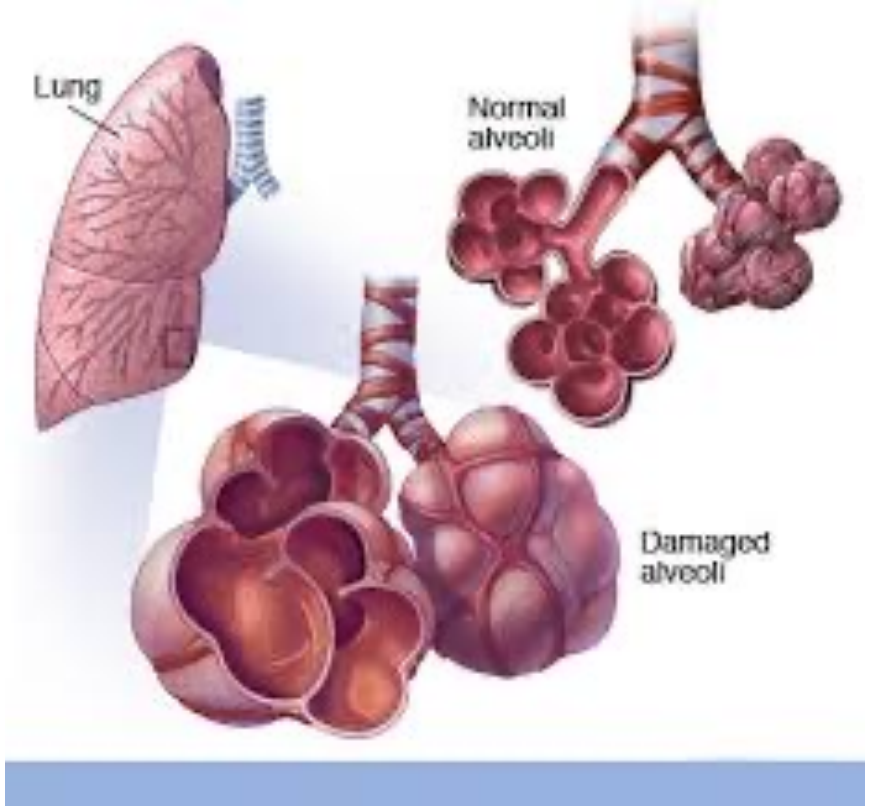


4. Surface Area

- Adult lung = approx. 300million alveoli
- If opened and spread out, would cover an entire tennis court
- Gas exchange SA = 70-80m_2

Clinical relevance:

- If SA ↓, despite adequate diffusion, sufficient gas exchange may not be possible resulting in: ↑ PCO_2 + ↓PO_2 + ↓pH
- Temp. loss of SA : Bronchial obstruction (mucus plug, tumor), actelectasis, consolidation
- Permanent loss of SA – emphysematous bullae (breakdown of alveolar walls)



5. Ventilation – Perfusion Coupling

- V = ventilation = air which reaches the lungs
- Q = perfusion = blood which reaches the lungs

Vital to maintain V/Q ratio to achieve adequate gas exchange

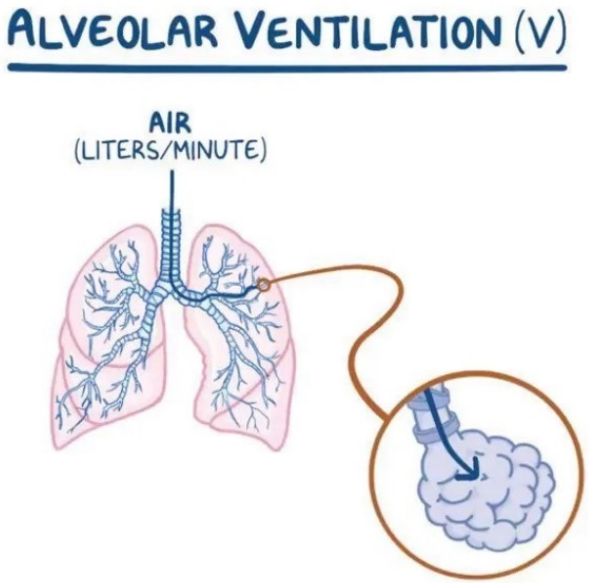
If off-balance = V/Q mismatch

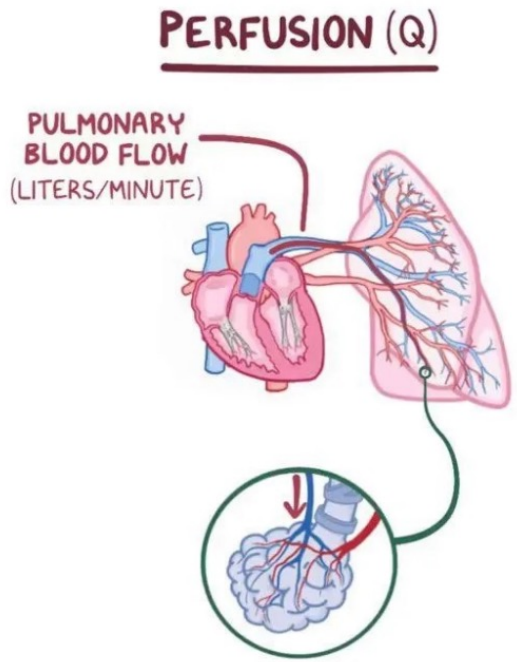
Alveolar dead space = ventilation but no perfusion

Eg: PE, blood clot, emphysema

Shunt = perfusion but no ventilation

Eg: Atelectasis, airway obstruction, pulmonary oedema



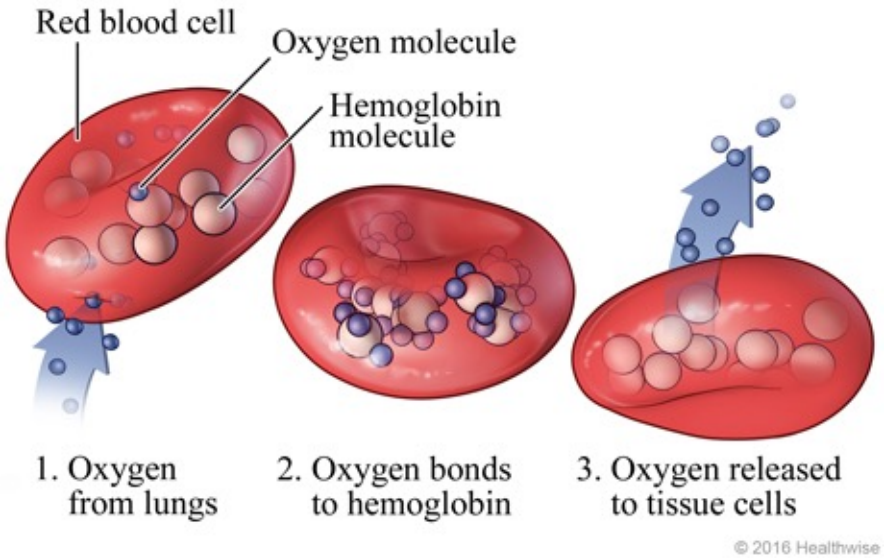


OXYGEN TRANSPORT:

O2 **does not easily dissolve in water** so needs a carrier haemoglobin (Hb) to be transported in blood

- 98.5% of blood O2 is bound to haemoglobin in RBC
  - Small diameter of the capillaries surrounding the alveoli - only one red blood cell can pass through at a time.
  - Slows down the blood cells - gases have more time to diffuse through
  - It also forces the red blood cells close to the walls of the capillaries, decreasing the diffusion distance.





TERMINOLOGY:

PO2: Partial pressure of oxygen = its concentration

PaO2: Partial pressure of oxygen in arterial blood (10.5 – 13.5 kPa / 82.5-105mmHg)

SaO2: Level of saturation of haemoglobin with oxygen, in arterial blood (95-98%)

SpO2: As above but described in terms of being measured by pulse oximetry

FiO2: Fraction (concentration) of inspired O2

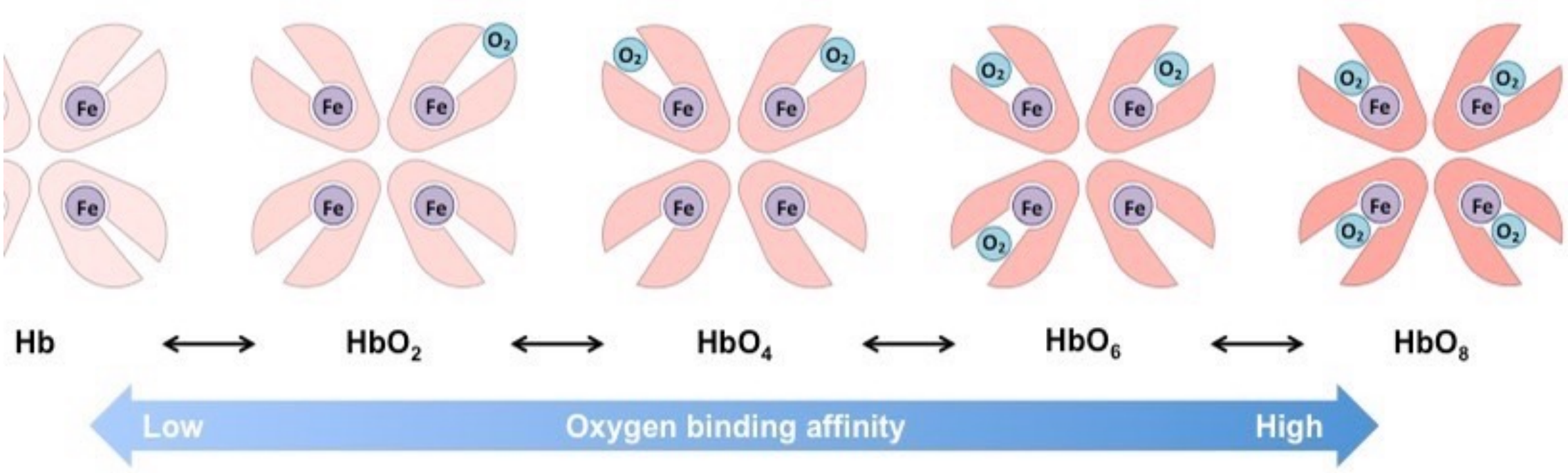
Haemoglobin (Hb): Main protein within RBCs binds to and transports O2 around the body

Affinity: strength by which two (or more) molecules interact or bind.

Dissociation : The act of separating or state of being separated.

Haemoglobin

- Haem portion contains 4 atoms of iron, each capable of binding to one O2 (oxyhaemoglobin HbO2)
- When O2 binds - Hb alters shape > from T shape to R shape and is cooperative
- Normally Hb is almost completely saturated in room air 96% (FiO2 = 21%)
- As blood flows through capillaries, haem - oxygen reaction reverses and Haemoglobin releases oxygen, which diffuses into interstitial fluid and then into cells



Hb's changing Affinity for O_2

- Maximises O_2 binding at lungs
- Facilitates O_2 release at tissues
- Produces sigmoid shape oxygen dissociation curve

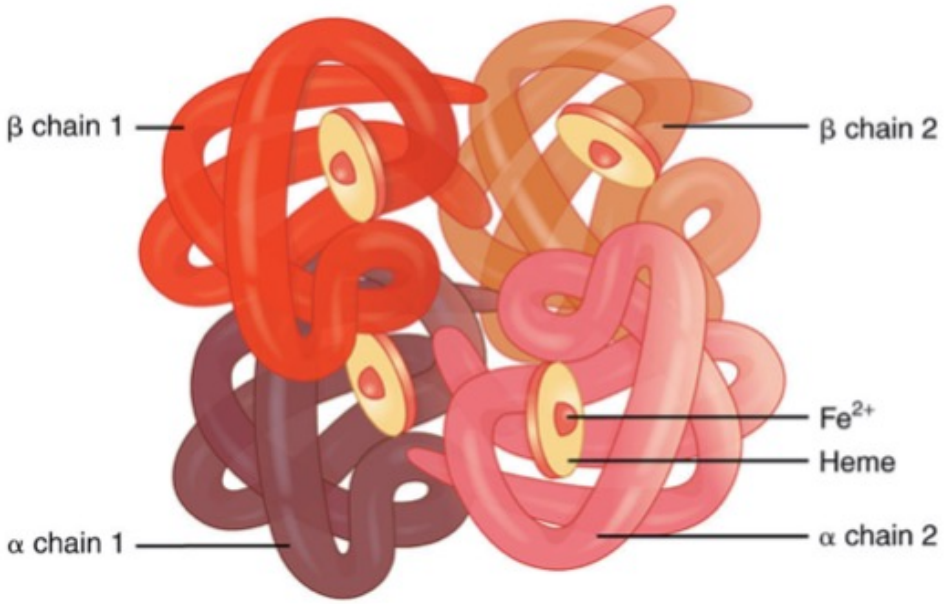
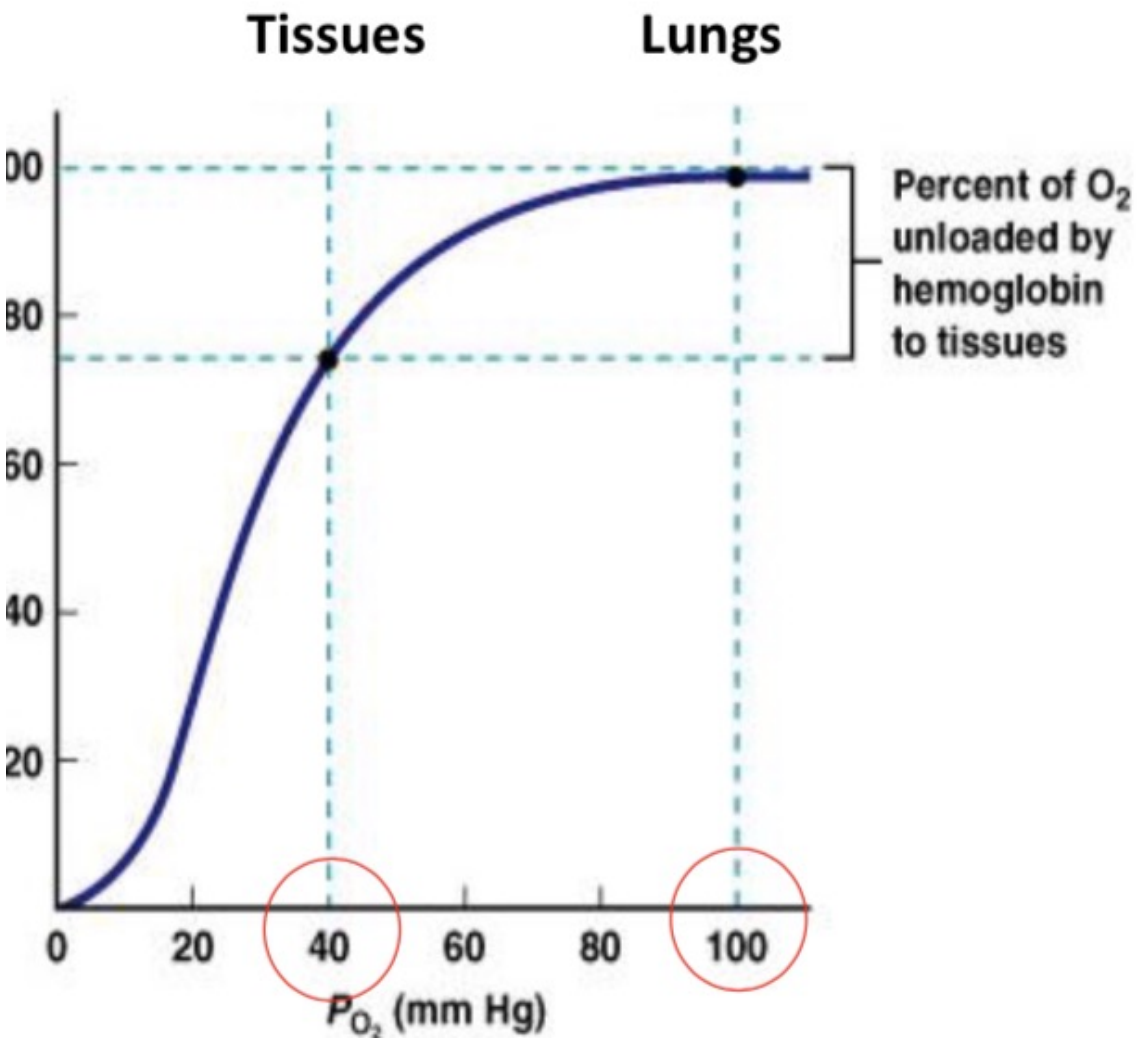


Fig 1. Structure of Haemoglobin.

O2 - Haemoglobin Dissociation Curve:

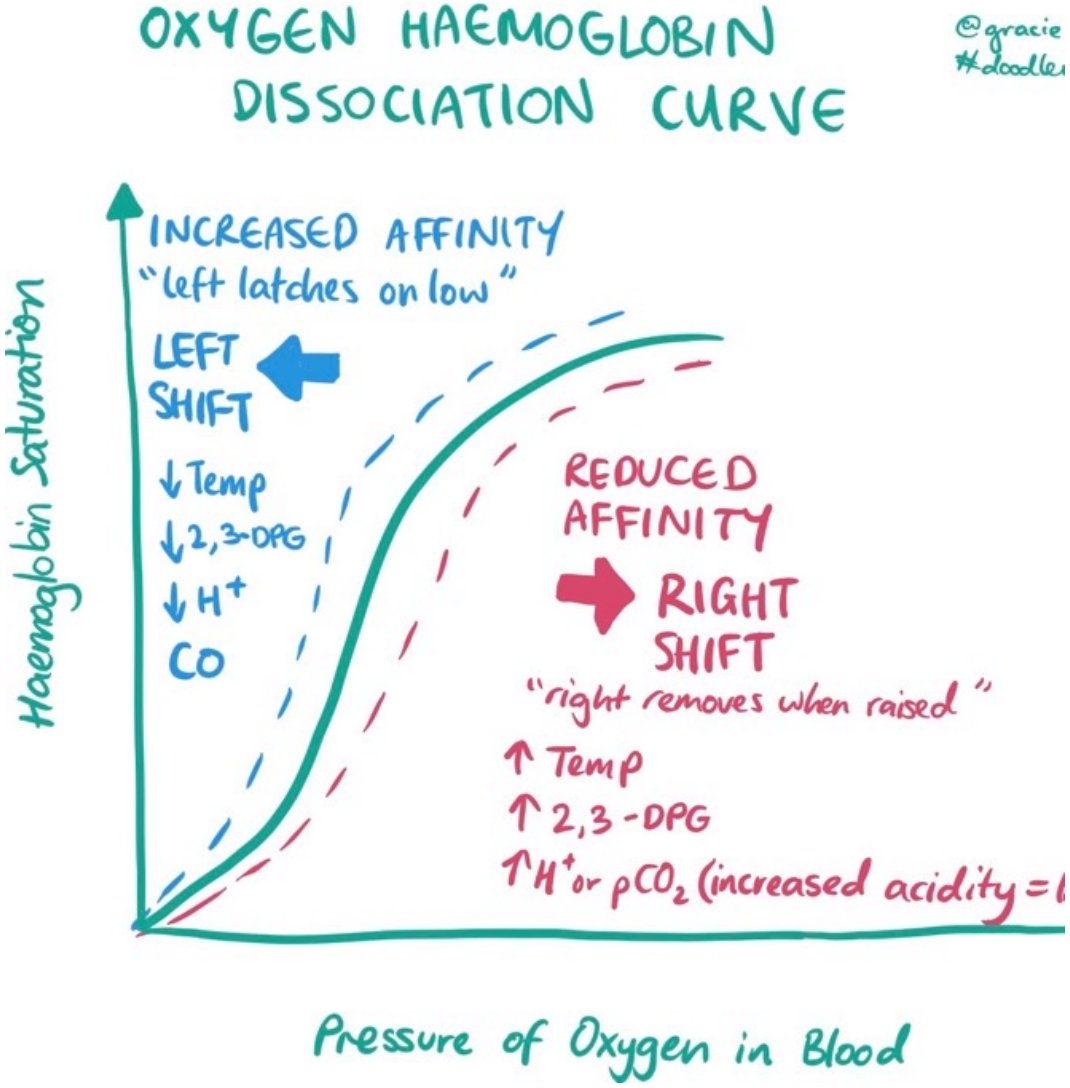
- At the lung, oxygen diffuses in to blood stream and binds to Hb (High SaO2 and PaO2)
- At the tissue oxygen dissociates from Hb to enter the tissue (Low SaO2 and PaO2)
- Steep section of curve is dissociation - results in large change in % saturation which allows for large quantities of O2 to be dumped in the tissues
- CO and NO have a higher affinity for Hb than O2 so they would shift curve to left - causes deoxygenation of tissues with normal PO2.

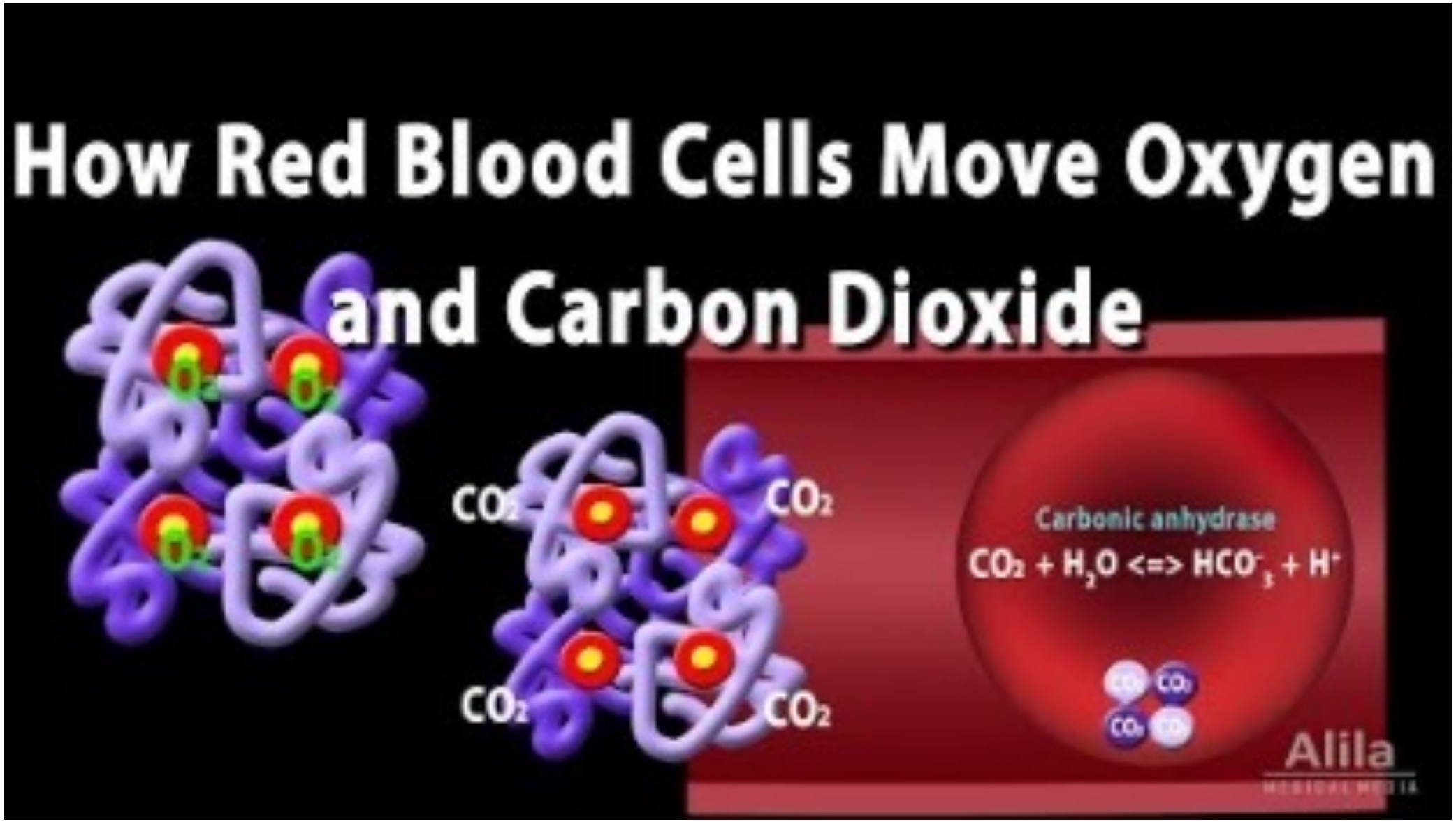


O2 - Haemoglobin Dissociation Curve:

- If the curve shifts to the RIGHT = increase release of O2 from Hb to tissue (give O2 to tissue)

- Shift curve to LEFT = more O2 on Hb so decreases release of O2 from Hb to tissue (less release of O2 to the tissue)




How Red Blood Cells Move Oxygen
and Carbon Dioxide
O_2
O_2
O_2
O_2
CO_2
CO_2
CO_2
CO_2
Carbonic anhydrase
$CO_2 + H_2O <=> HCO^-_3 + H^+$
Alila

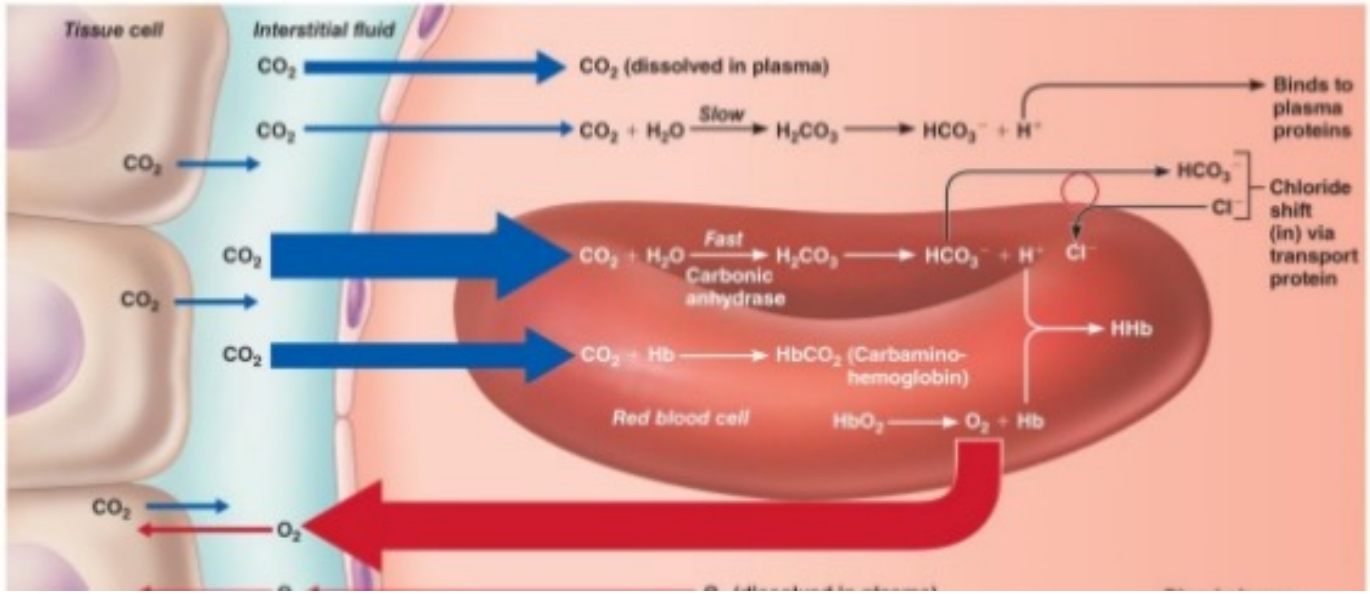
Considerations of CO2 transport

CO_2 Transport

$$CO_2 + H_2O \rightleftharpoons H_2CO_3 \rightleftharpoons H^+ + HCO_3^-$$

carbon dioxide — water — carbonic acid — hydrogen ion — bicarbonate ion

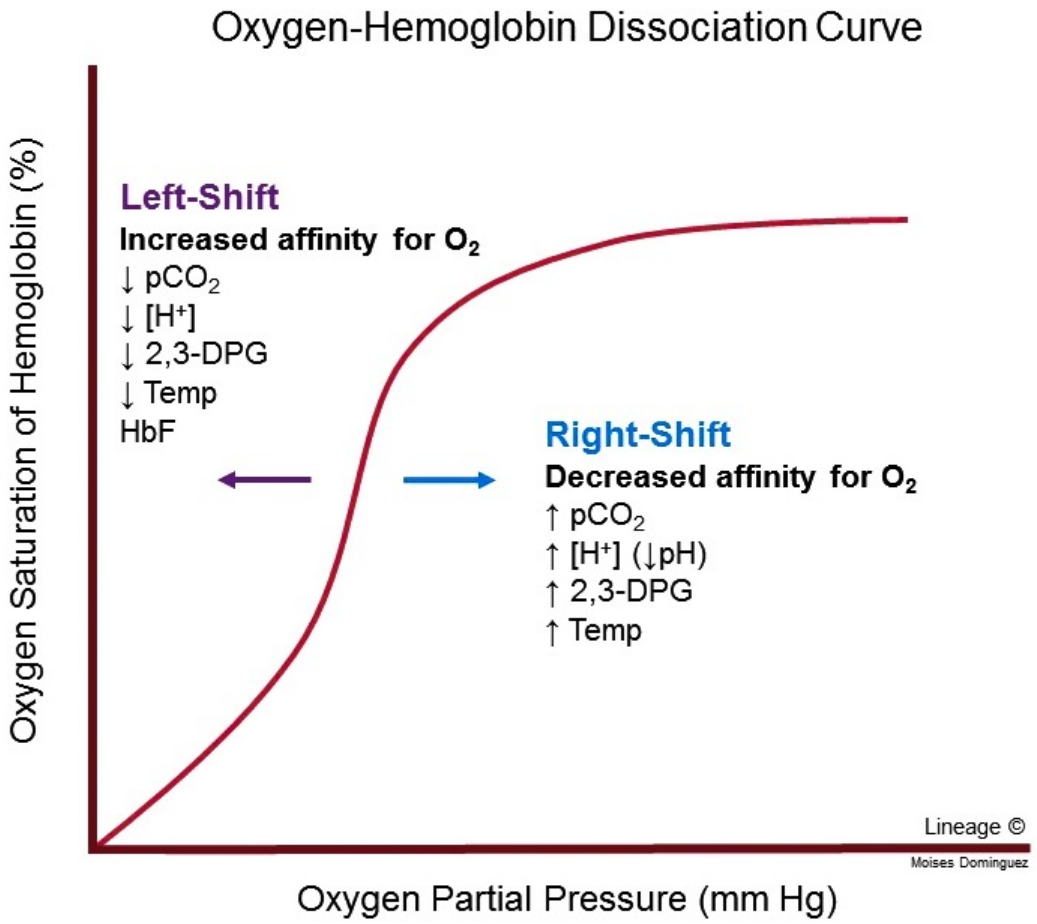
- CO_2 transported in blood in three forms
  - 7 to 10% dissolved in plasma
  - 20% bound to *globin* of hemoglobin (**carbaminohemoglobin**)
  - 70% transported as **bicarbonate ions** (HCO_3^-) in plasma
- CO_2 combines with water to form carbonic acid (H_2CO_3), which quickly dissociates
- Occurs primarily in RBCs, where **carbonic anhydrase** reversibly and rapidly catalyzes reaction



Haemoglobin and the Bohr Effect



FACTORS AFFECTING AFFINITY OF HAEMOGLOBIN:



CADET shift RIGHT

C = CO2

A = Acidity

D = DPG (BPG)

E = Exercise

T = Temperature





THANKS FOR LISTENING



QUESTIONS